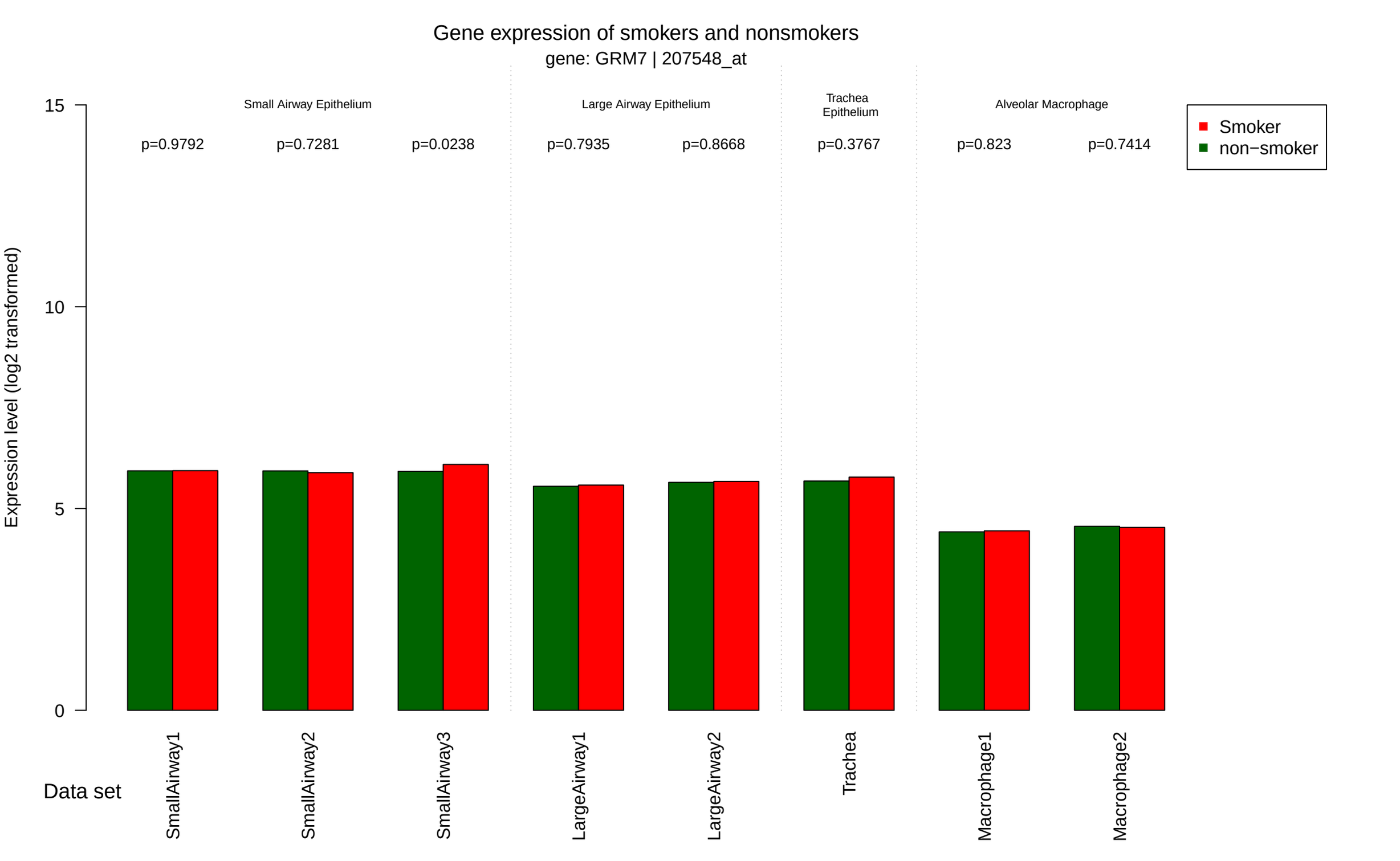
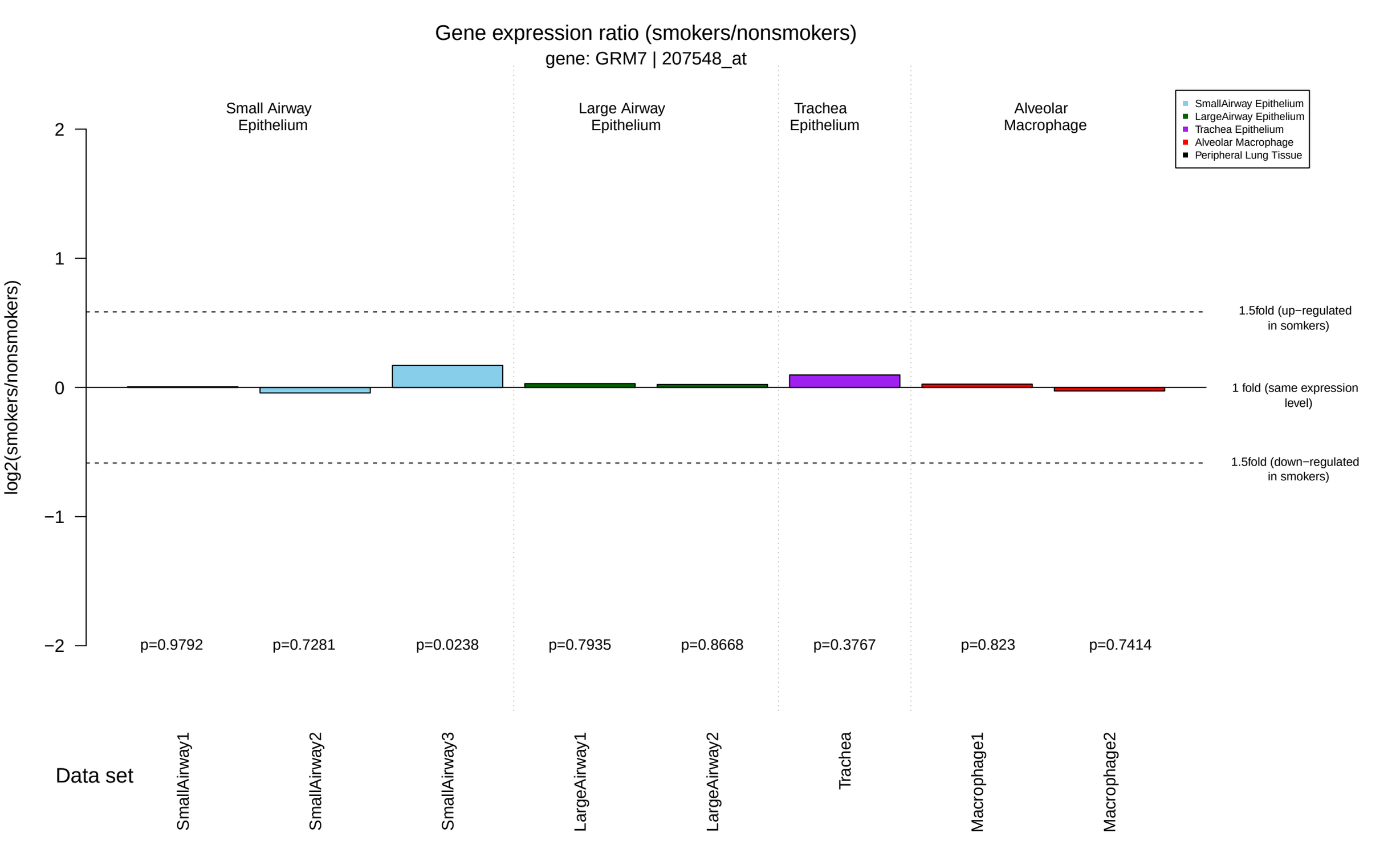
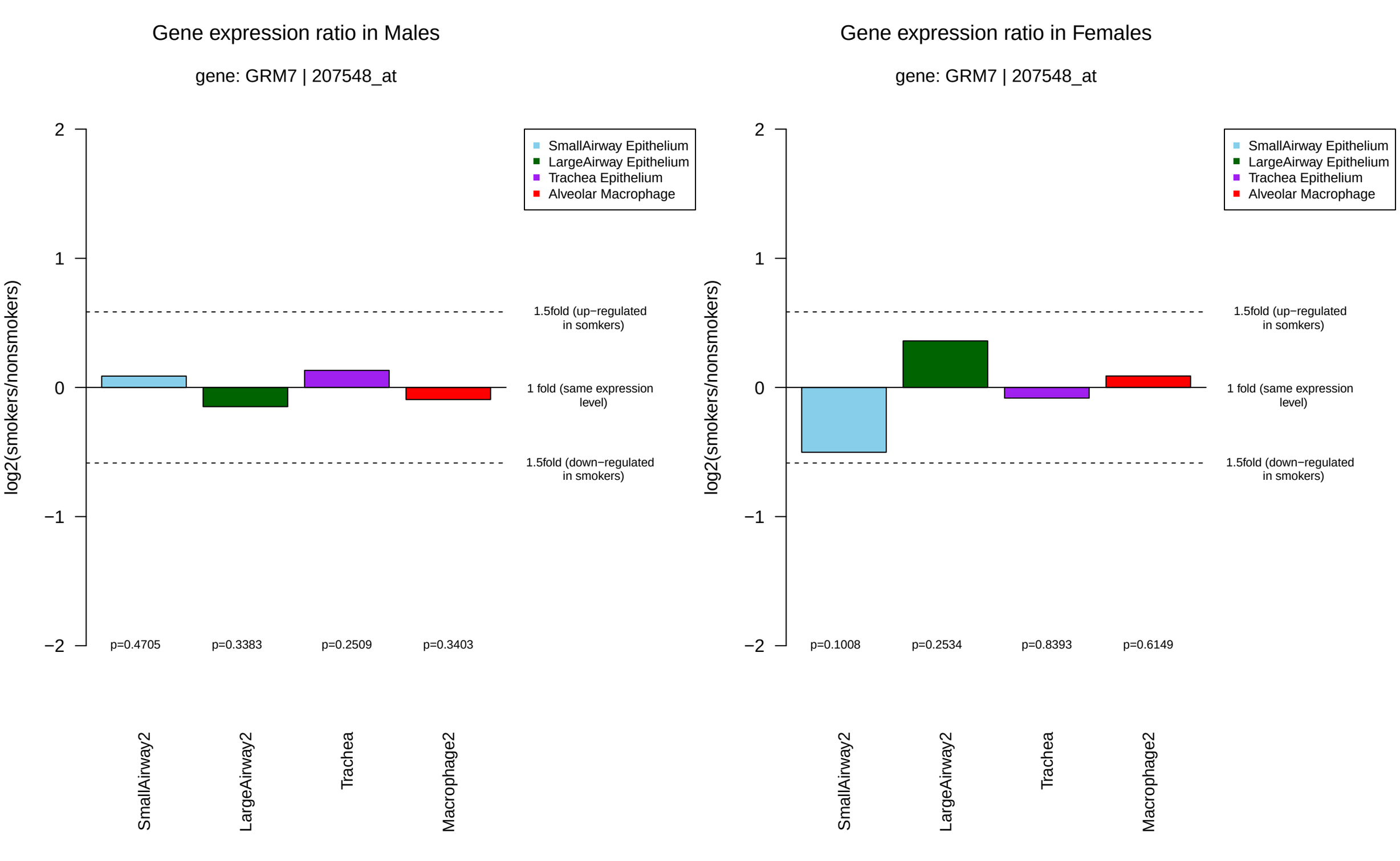
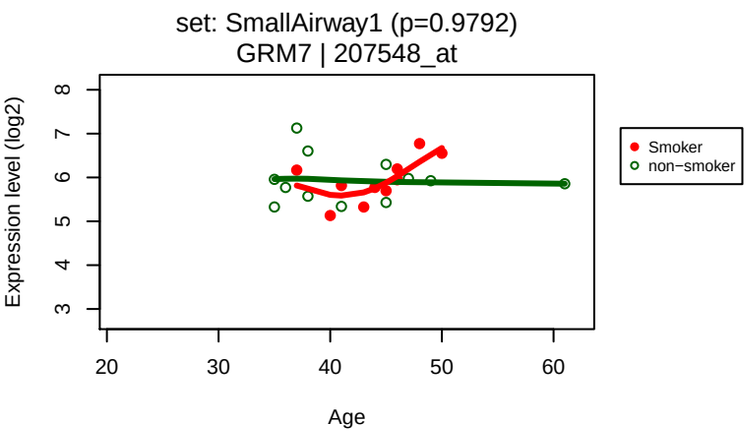
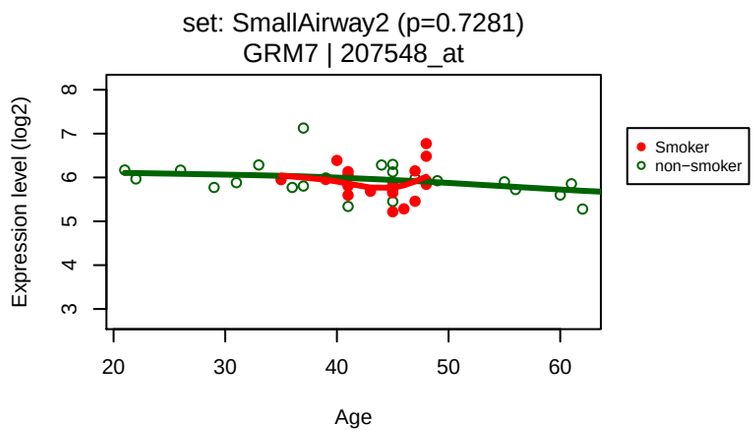
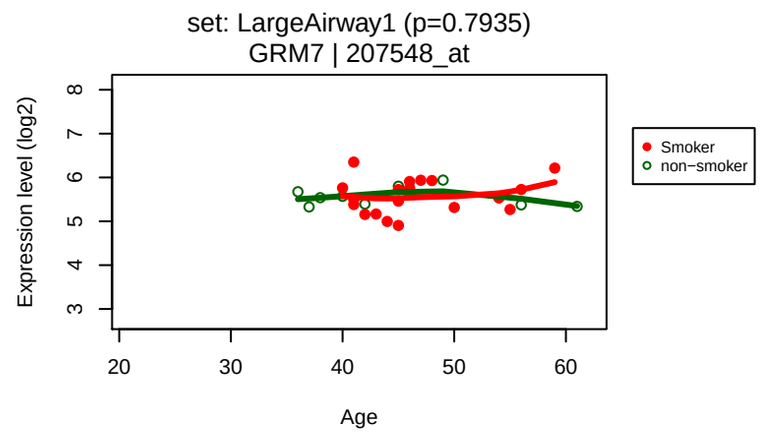

Gene expression of smokers and nonsmokers
gene: GRM7 | 207548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9792
p=0.7281
p=0.0238
p=0.7935
p=0.8668
p=0.3767
p=0.823
p=0.7414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

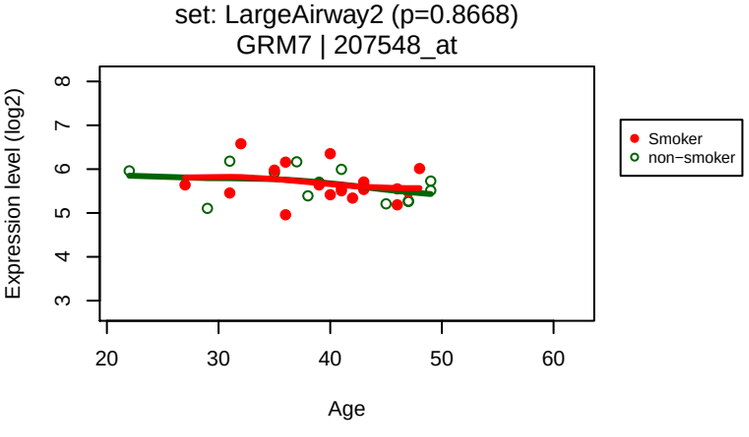

Gene expression ratio (smokers/nonsmokers)
gene: GRM7 | 207548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9792
p=0.7281
p=0.0238
p=0.7935
p=0.8668
p=0.3767
p=0.823
p=0.7414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

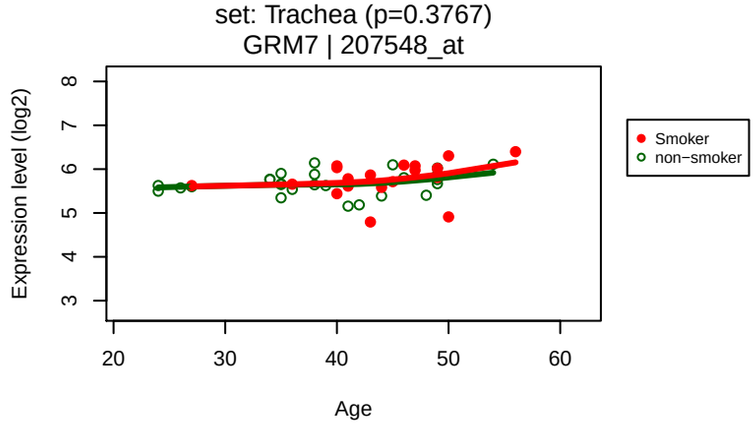

Gene expression ratio in Males
gene: GRM7 | 207548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4705
p=0.3383
p=0.2509
p=0.3403
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM7 | 207548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1008
p=0.2534
p=0.8393
p=0.6149
SmallAirway2
LargeAirway2
Trachea
Macrophage2

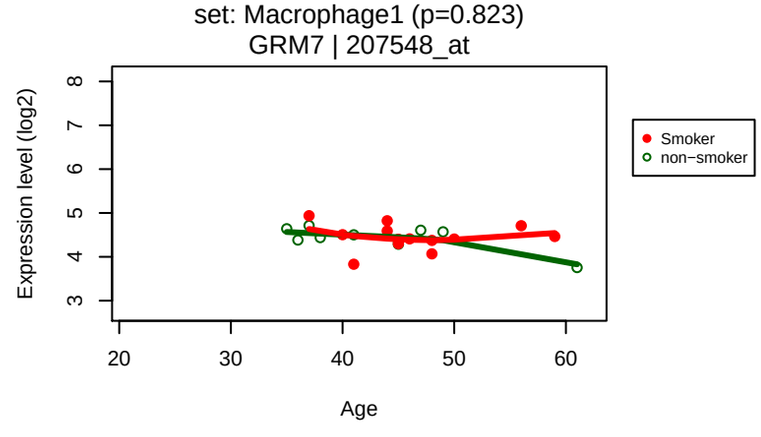

set: SmallAirway1 (p=0.9792)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker

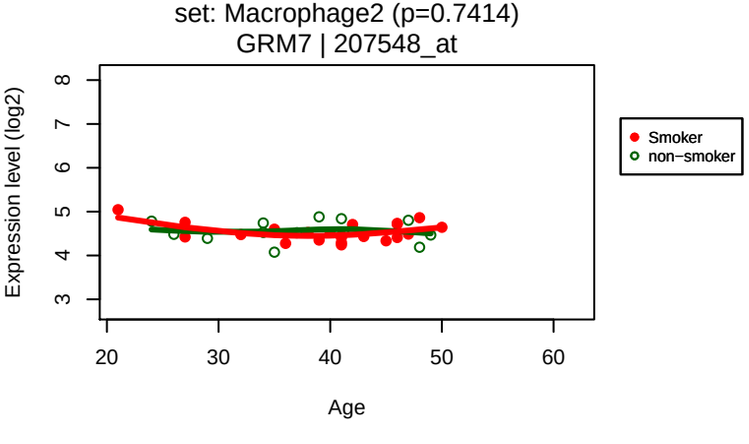

set: SmallAirway2 (p=0.7281)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker

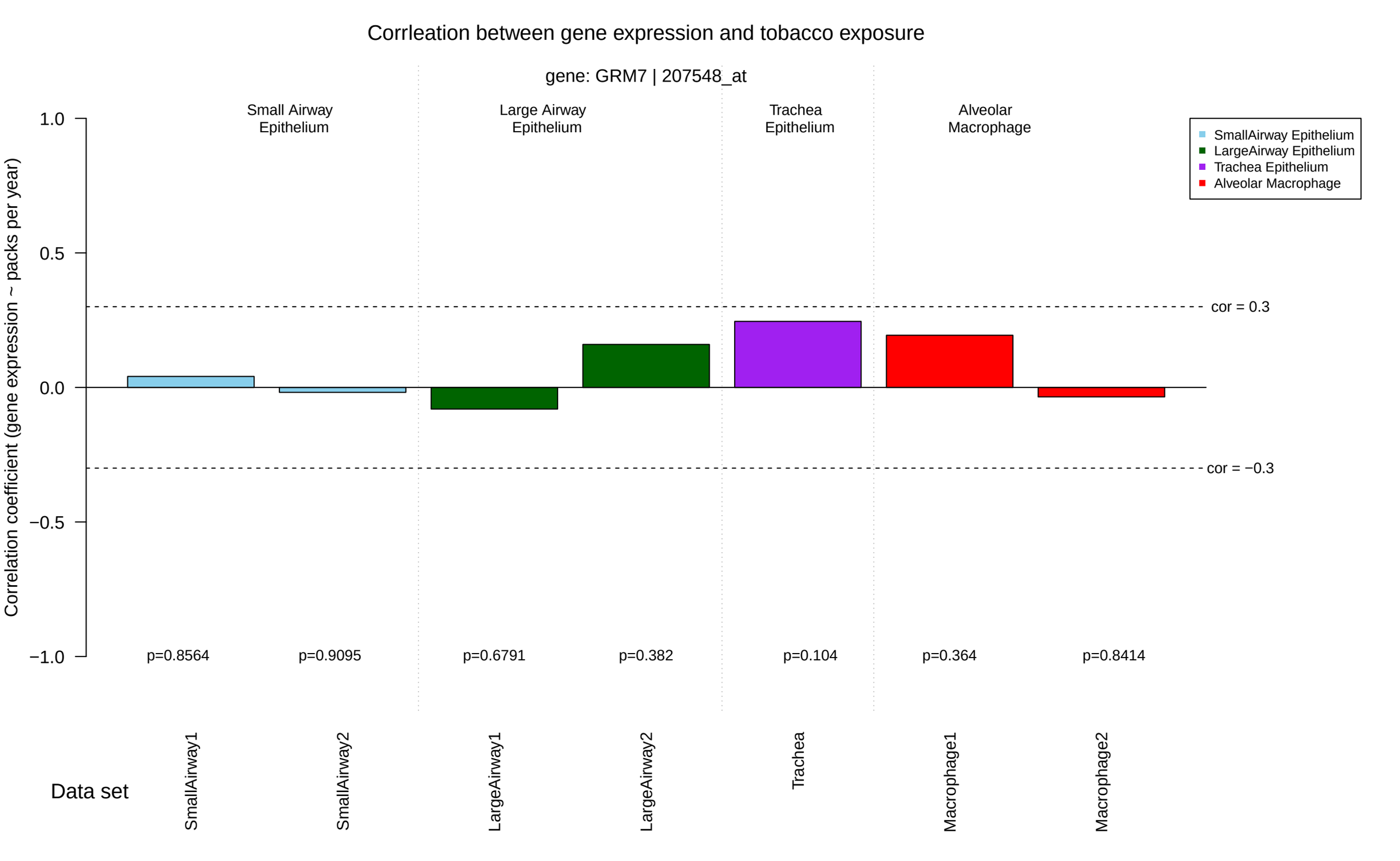

set: LargeAirway1 (p=0.7935)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8668)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3767)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.823)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7414)
GRM7 | 207548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM7 | 207548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8564
p=0.9095
p=0.6791
p=0.382
p=0.104
p=0.364
p=0.8414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2